

Stabilizing Quantities ρ_p for Boundary Control of PDE: Proof of Conjecture 3

Based on a question by Jean-Michel Coron (2016)
Original problem statement by Amaury Hayat (February 09, 2026)

Setting and Definitions

We consider nonlinear PDE of the form

$$\partial_t u + A(u)\partial_x u + B(u) = 0, \quad \text{on } [0, L],$$

where $u \in \mathbb{R}^n$, with boundary conditions

$$\begin{pmatrix} u_+(t, 0) \\ u_-(t, L) \end{pmatrix} = G \begin{pmatrix} u_+(t, L) \\ u_-(t, 0) \end{pmatrix}.$$

A natural quantity arising in the stability analysis is the following function. For $M \in \mathbb{R}^{n \times n}$ and $p \in [1, +\infty]$, define

$$\rho_p(M) := \inf \{ \|\Delta M \Delta^{-1}\|_p \mid \Delta = \text{diag}(\delta_1, \dots, \delta_n), \delta_i > 0 \forall i \},$$

where $\|\cdot\|_p$ denotes the matrix p -norm (operator norm induced by the vector ℓ^p norm).

Background

It is known that for any $n \in \mathbb{N}$ and $M \in \mathbb{R}^{n \times n}$:

$$\rho_1(M) \geq \rho_2(M), \quad \rho_1(M) = \rho_\infty(M),$$

and for $n \geq 2$ there exists $M \in \mathbb{R}^{n \times n}$ such that $\rho_1(M) > \rho_2(M)$. Moreover, if M has only non-negative coefficients then $\rho_k(M) = \rho_1(M)$ for all $k \in [1, +\infty]$.

The Conjectures

Conjecture 1 (Conjecture 1). *For any $M \in \mathbb{R}^{n \times n}$,*

$$\rho_2(M) \leq \rho_3(M) \leq \dots \leq \rho_\infty(M).$$

Conjecture 2 (Conjecture 2). *For any $M \in \mathbb{R}^{n \times n}$,*

$$\rho_2(M) \leq \rho_k(M), \quad \forall k \in \mathbb{N} \setminus \{0\}.$$

Conjecture 3 (Conjecture 3 — **To be proved**). *For any $n > 6$ and any $p > 2$ (with $n, p \in \mathbb{N}$), there exists $M \in \mathbb{R}^{n \times n}$ such that*

$$\rho_p(M) < \rho_{p+1}(M).$$

Goal

Prove Conjecture 3: for every integer $n > 6$ and every integer $p > 2$, exhibit an $n \times n$ matrix M (or a family thereof) for which $\rho_p(M) < \rho_{p+1}(M)$.

Note on the matrix p -norm. Recall that for a matrix $A \in \mathbb{R}^{n \times n}$,

$$\|A\|_p = \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p},$$

with $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ for $p < \infty$ and $\|x\|_\infty = \max_i |x_i|$. Special cases: $\|A\|_1 = \max_j \sum_i |a_{ij}|$ (max column sum), $\|A\|_\infty = \max_i \sum_j |a_{ij}|$ (max row sum), and $\|A\|_2 = \sigma_{\max}(A)$ (largest singular value).

References

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